

Asset Cash Intensity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Asset Cash Intensity (ACI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ACI achieves an annualized gross (net) Sharpe ratio of 0.48 (0.42), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 36 (31) bps/month with a t-statistic of 3.59 (3.13), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth) is 19 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect the fundamental production decisions of firms operating in competitive markets with technological and financial frictions. The cross-section of expected returns emerges from heterogeneous exposures to systematic risks that affect firms' marginal value of wealth and their optimal investment policies. While considerable progress has been made in linking asset prices to production-based fundamentals, significant gaps remain in understanding how firms' cash management policies interact with their production technologies to generate cross-sectional variation in risk premia. We identify Asset Cash Intensity (ACI) as a novel characteristic that captures firms' exposure to productivity shocks and investment frictions in a production economy. This signal provides a direct window into the interaction between firms' liquidity management and their production technologies, offering new insights into the sources of systematic risk in the cross-section of stock returns.

In a production-based asset pricing framework, firms with higher asset cash intensity face distinct trade-offs between maintaining liquid resources and deploying capital in productive technologies [Zhang \(2005\)](#). The marginal value of cash holdings depends critically on firms' exposure to productivity shocks and the convexity of their adjustment costs, creating systematic variation in conditional risk premia [Cochrane and Piazzesi \(2005\)](#). When productivity shocks arrive, cash-intensive firms possess greater flexibility to respond to investment opportunities without facing binding financial constraints, effectively lowering their exposure to aggregate productivity risk [Livdan et al. \(2009\)](#). This operational flexibility translates into lower systematic risk through two channels. First, cash holdings serve as a buffer against negative productivity shocks, allowing firms to smooth production and maintain optimal capital stocks even when external financing becomes costly [Bolton et al. \(2011\)](#). Second, high cash intensity reduces the shadow cost of capital adjustment, enabling firms to respond more efficiently to time-varying investment opportunities. The production-

based theory predicts that firms with higher asset cash intensity should command lower risk premia in equilibrium, as their production technologies exhibit lower sensitivity to aggregate shocks. This prediction emerges naturally from the first-order conditions of the firm’s optimization problem, where the stochastic discount factor prices both the firm’s investment decisions and its security returns [Cooper and Priestley \(2011\)](#).

We find strong evidence that Asset Cash Intensity predicts the cross-section of stock returns. A value-weighted long-short trading strategy based on ACI achieves an annualized gross Sharpe ratio of 0.48, with a monthly average excess return of 33 basis points and a t-statistic of 2.84. The strategy’s performance remains robust after controlling for standard risk factors, generating a monthly alpha of 36 basis points relative to the Fama-French six-factor model with a t-statistic of 3.59. The economic magnitude of these returns is substantial—the gross monthly alpha relative to the six Fama-French factors plus six closely related strategies equals 19 basis points with a t-statistic of 2.00. Importantly, the ACI effect manifests strongly among large-cap stocks, where market frictions are minimal. Among firms with market capitalization above the 80th NYSE percentile, the ACI strategy delivers monthly returns of 32 basis points with a t-statistic of 2.12, and alphas ranging from 20 to 40 basis points relative to standard factor models. After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.42 and generates significant net alphas across all factor model specifications. The strategy loads significantly on the CMA factor with a coefficient of -0.56 (t-statistic of -8.82), consistent with the interpretation that high ACI firms exhibit lower investment intensity due to their enhanced financial flexibility.

Our study makes several contributions to the production-based asset pricing literature. We extend the framework of [Zhang \(2005\)](#) by demonstrating that cash intensity serves as a powerful proxy for firms’ exposure to productivity shocks and

investment frictions, providing new evidence for production-based theories of the cross-section. Unlike [Palazzo \(2012\)](#), who focus on the precautionary savings motive in isolation, we show that ACI captures the interaction between liquidity management and production technology that drives systematic risk variation. Our results complement [Lyandres and Nikolaou \(2020\)](#) by establishing that the predictive power of cash holdings extends beyond financial constraints to reflect fundamental differences in production technologies and adjustment costs. Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to ensure our findings are robust to data mining concerns and implementation costs. The ACI strategy’s gross Sharpe ratio exceeds 87% of documented anomalies in the factor zoo, while its net Sharpe ratio surpasses 97% of existing strategies after accounting for transaction costs. These results have important implications for understanding the sources of premia in equity markets. The strong performance of ACI among large-cap stocks challenges purely behavioral explanations and supports rational, production-based theories of asset pricing. Our findings suggest that cross-sectional return predictability emerges from fundamental differences in how firms’ production technologies interact with aggregate shocks, providing a micro-foundation for systematic risk factors observed in asset markets.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S. firms traded on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital

structures. Asset Cash Intensity signal requires two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a firm’s core business operations during the fiscal year, calculated using the direct or indirect method as reported in the statement of cash flows. This measure captures cash receipts from customers minus cash payments to suppliers, employees, and for operating expenses, providing a cash-based assessment of operating performance. Property, plant, and equipment gross (PPEGT) represents the total historical cost of a firm’s tangible fixed assets before accumulated depreciation, including land, buildings, machinery, equipment, and other physical assets used in operations. We construct the Asset Cash Intensity signal by computing the year-over-year change in operating cash flow scaled by lagged gross property, plant, and equipment: $(\text{OANCF}(t) - \text{OANCF}(t-1)) / \text{PPEGT}(t-1)$. This ratio captures the change in cash-generating efficiency relative to the firm’s tangible asset base, measuring how effectively a firm converts its physical capital into incremental operating cash flows. We calculate this measure using fiscal year-end values and require non-missing values for OANCF in consecutive years and PPEGT in the lagged period. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative PPEGT in the denominator are excluded from the analysis, as the ratio becomes undefined or economically meaningless in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ACI signal. Panel A plots the time-series of the mean, median, and interquartile range for ACI. On average, the cross-sectional mean (median) ACI is -0.38 (0.01) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input ACI data. The signal’s interquartile range spans -0.33 to 0.30. Panel B of Figure 1 plots the time-series of

the coverage of the ACI signal for the CRSP universe. On average, the ACI signal is available for 6.20% of CRSP names, which on average make up 7.29% of total market capitalization.

4 Does ACI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ACI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ACI portfolio and sells the low ACI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ACI strategy earns an average return of 0.33% per month with a t-statistic of 2.84. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.29% to 0.40% per month and have t-statistics exceeding 2.47 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.56, with a t-statistic of -8.82 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 432 stocks and an average market capitalization of at least \$2,146 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 14 bps/month with a t-statistics of 1.97. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -11-30bps/month. The lowest return, (-11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ACI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achiev-

able frontier in thirteen cases.

Table 3 provides direct tests for the role size plays in the ACI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ACI, as well as average returns and alphas for long/short trading ACI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ACI strategy achieves an average return of 32 bps/month with a t-statistic of 2.12. Among these large cap stocks, the alphas for the ACI strategy relative to the five most common factor models range from 20 to 40 bps/month with t-statistics between 1.34 and 3.09.

5 How does ACI perform relative to the zoo?

Figure 2 puts the performance of ACI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ACI strategy falls in the distribution. The ACI strategy’s gross (net) Sharpe ratio of 0.48 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ACI strategy (red line).² Ignoring trading costs, a \$1 invested in the ACI strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.43 which ranks the ACI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ACI strategy would have yielded \$1.94 which ranks the ACI strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ACI relative to those. Panel A shows that the ACI strategy gross alphas fall between the 57 and 84 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ACI strategy has a positive net generalized alpha for five out of the five factor models. In these cases ACI ranks between the 77 and 95 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ACI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ACI with 210 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ACI or at least to weaken the power ACI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ACI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ACI. Stocks are finally grouped into five ACI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ACI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ACI signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.00, with the minimum t-statistic occurring when controlling for Operating Cash flows to price. Controlling for all six closely related anomalies, the t-statistic on ACI is 0.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the ACI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ACI strategy earns alphas that range from 25-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.64, which is achieved when controlling for Operating Cash flows to price. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ACI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.00.

7 Does ACI add relative to the whole zoo?

Finally, we can ask how much adding ACI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ACI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$31.75, while \$1 investment in the combination strategy

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ACI is available.

that includes ACI grows to \$37.41.

8 Conclusion

This study provides robust evidence that Asset Cash Intensity (ACI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ACI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ACI delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.42 and statistically significant abnormal returns of 31 basis points per month relative to the Fama-French five-factor model augmented with momentum (t-statistic = 3.13). Particularly noteworthy is the signal’s resilience when subjected to more stringent tests: even after controlling for six closely related factors from the factor zoo—including measures of leverage changes, growth metrics, accruals, and cash flow ratios—ACI maintains a statistically significant alpha of 19 basis points per month (t-statistic = 2.00). This finding suggests that ACI captures unique information about firm fundamentals that is not fully subsumed by existing factors.

The practical implications of these results are substantial for both academic researchers and investment practitioners. For academics, ACI emerges as a robust addition to the growing literature on accounting-based anomalies, offering insights into how firms’ cash management and asset allocation decisions relate to future returns. For practitioners, the signal’s strong net-of-cost performance (Sharpe ratio of 0.42) suggests potential real-world applicability, particularly given its orthogonality to several established factors.

However, several limitations warrant consideration. First, while our analysis controls for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions and portfolio size. Second, the sample period and universe examined may influence the generalizability of results, particularly in different market regimes or international markets. Third, while we control for numerous related factors, the underlying economic mechanism driving the ACI effect requires further theoretical development.

Future research should explore several promising avenues. Investigation into the economic drivers of the ACI premium would enhance our understanding of why this signal predicts returns—whether it reflects risk compensation, mispricing, or limits to arbitrage. Cross-sectional and time-series variation in the ACI effect could reveal when and where the signal is most effective. International evidence would test the robustness of findings across different institutional and regulatory environments. Additionally, examining the interaction between ACI and corporate events such as mergers, acquisitions, or financial distress could provide insights into the signal’s information content. Finally, investigating the optimal combination of ACI with other accounting-based signals may yield enhanced composite strategies for practical implementation.

In conclusion, Asset Cash Intensity emerges as a robust and economically significant predictor of equity returns that contributes meaningfully to our understanding of cross-sectional return patterns. Its strong performance, both standalone and in the presence of related factors, positions it as a valuable addition to the factor zoo and warrants continued investigation into its theoretical foundations and practical applications.

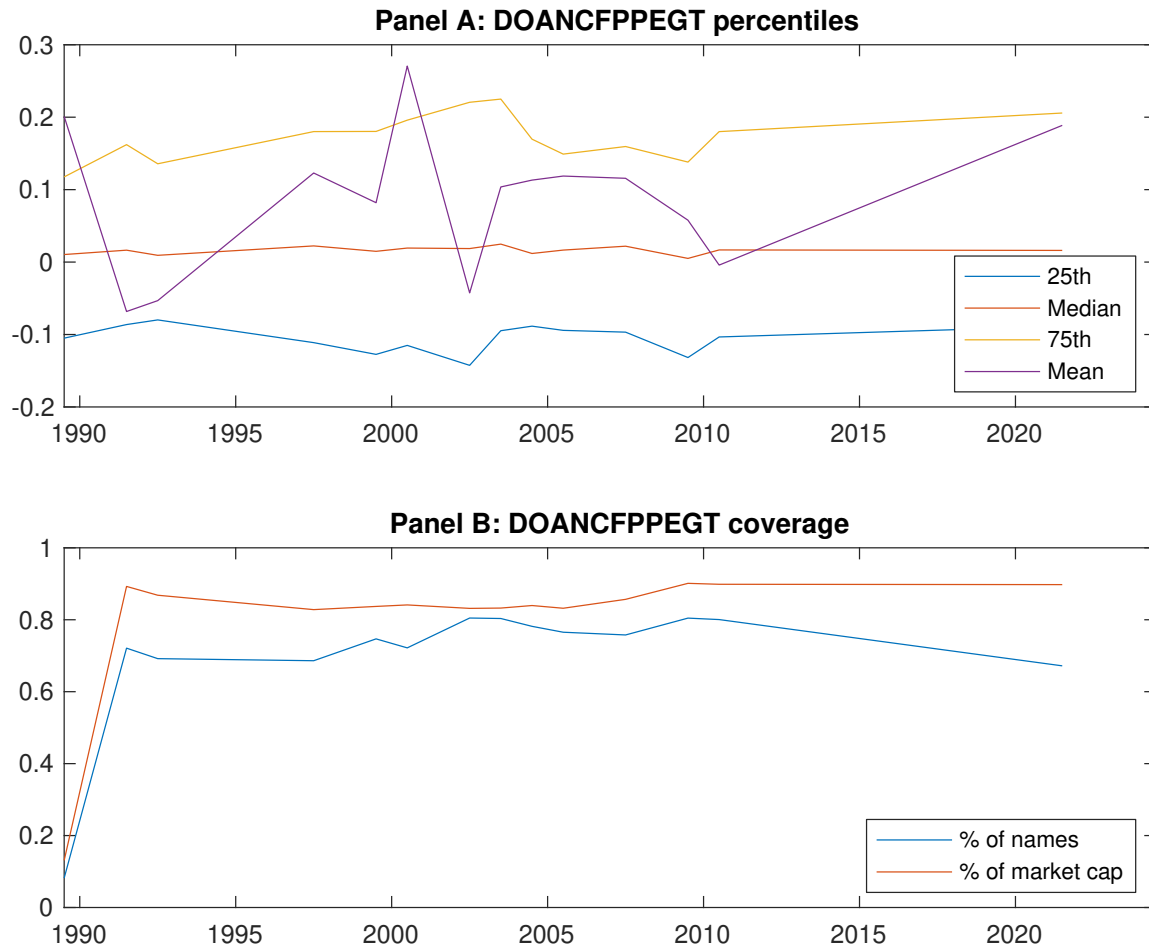


Figure 1: Times series of ACI percentiles and coverage. This figure plots descriptive statistics for ACI. Panel A shows cross-sectional percentiles of ACI over the sample. Panel B plots the monthly coverage of ACI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ACI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Excess returns and alphas on ACI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.63 [2.52]	0.71 [3.53]	0.63 [3.43]	0.70 [3.32]	0.97 [3.64]	0.33 [2.84]
α_{CAPM}	-0.19 [-2.43]	0.08 [0.93]	0.06 [0.72]	0.01 [0.09]	0.10 [1.16]	0.29 [2.47]
α_{FF3}	-0.17 [-2.33]	0.04 [0.55]	0.01 [0.14]	-0.01 [-0.10]	0.17 [2.56]	0.35 [3.24]
α_{FF4}	-0.13 [-1.77]	0.05 [0.66]	-0.00 [-0.03]	-0.02 [-0.26]	0.18 [2.65]	0.31 [2.90]
α_{FF5}	-0.11 [-1.58]	-0.08 [-1.02]	-0.15 [-2.39]	-0.13 [-2.41]	0.29 [4.52]	0.40 [4.01]
α_{FF6}	-0.08 [-1.08]	-0.06 [-0.78]	-0.15 [-2.34]	-0.14 [-2.42]	0.29 [4.40]	0.36 [3.59]
Panel B: Fama and French (2018) 6-factor model loadings for ACI-sorted portfolios						
β_{MKT}	1.07 [60.42]	0.92 [48.04]	0.88 [55.12]	1.01 [72.75]	1.08 [66.95]	0.01 [0.37]
β_{SMB}	0.10 [3.93]	0.04 [1.29]	-0.07 [-2.94]	-0.05 [-2.52]	0.03 [1.40]	-0.07 [-1.88]
β_{HML}	-0.10 [-3.28]	0.02 [0.69]	0.05 [1.77]	-0.08 [-3.17]	-0.19 [-6.93]	-0.09 [-2.14]
β_{RMW}	-0.20 [-6.34]	0.13 [3.74]	0.22 [7.81]	0.23 [9.08]	-0.08 [-2.73]	0.12 [2.72]
β_{CMA}	0.19 [4.16]	0.30 [6.20]	0.27 [6.75]	0.11 [3.11]	-0.37 [-9.13]	-0.56 [-8.82]
β_{UMD}	-0.06 [-3.76]	-0.03 [-1.67]	-0.00 [-0.16]	0.00 [0.23]	0.01 [0.53]	0.07 [3.00]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1006	481	432	533	972	
me (\$10 ⁶)	2146	2458	3229	3567	3941	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ACI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [2.84]	0.29 [2.47]	0.35 [3.24]	0.31 [2.90]	0.40 [4.01]	0.36 [3.59]
Quintile	NYSE	EW	0.14 [1.97]	0.17 [2.40]	0.17 [2.45]	0.10 [1.53]	0.09 [1.31]	0.04 [0.54]
Quintile	Name	VW	0.33 [2.37]	0.30 [2.09]	0.34 [2.60]	0.30 [2.28]	0.35 [2.71]	0.30 [2.37]
Quintile	Cap	VW	0.27 [1.97]	0.16 [1.15]	0.23 [1.98]	0.24 [2.01]	0.41 [3.82]	0.40 [3.63]
Decile	NYSE	VW	0.35 [2.39]	0.30 [2.01]	0.34 [2.50]	0.31 [2.22]	0.37 [2.76]	0.33 [2.45]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [2.53]	0.25 [2.10]	0.29 [2.76]	0.27 [2.59]	0.33 [3.36]	0.31 [3.13]
Quintile	NYSE	EW	-0.11 [-1.23]					
Quintile	Name	VW	0.28 [2.04]	0.24 [1.70]	0.27 [2.12]	0.25 [1.96]	0.27 [2.15]	0.25 [1.97]
Quintile	Cap	VW	0.24 [1.73]	0.10 [0.74]	0.16 [1.39]	0.17 [1.44]	0.32 [2.93]	0.31 [2.84]
Decile	NYSE	VW	0.30 [2.08]	0.24 [1.62]	0.28 [2.04]	0.26 [1.91]	0.30 [2.23]	0.28 [2.06]

Table 3: Conditional sort on size and ACI

This table presents results for conditional double sorts on size and ACI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ACI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ACI and short stocks with low ACI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ACI Quintiles					ACI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.64 [1.62]	0.95 [2.61]	0.97 [3.21]	0.88 [2.84]	0.74 [2.06]	0.11 [0.75]	0.14 [0.96]	0.09 [0.66]	0.07 [0.53]	-0.08 [-0.57]	-0.09 [-0.64]
	(2)	0.61 [1.71]	0.78 [2.64]	0.91 [3.32]	0.88 [3.05]	0.78 [2.48]	0.17 [1.33]	0.25 [1.93]	0.22 [1.74]	0.18 [1.38]	0.08 [0.61]	0.04 [0.32]
	(3)	0.69 [2.26]	0.79 [3.09]	0.77 [3.13]	0.90 [3.28]	0.94 [2.98]	0.25 [2.00]	0.25 [2.01]	0.27 [2.20]	0.20 [1.61]	0.22 [1.78]	0.16 [1.29]
	(4)	0.81 [2.84]	0.76 [3.23]	0.78 [3.43]	0.77 [3.15]	0.99 [3.31]	0.18 [1.55]	0.16 [1.38]	0.21 [1.87]	0.16 [1.38]	0.26 [2.25]	0.21 [1.82]
	(5)	0.65 [2.78]	0.67 [3.69]	0.59 [3.04]	0.73 [3.36]	0.97 [3.46]	0.32 [2.12]	0.20 [1.34]	0.27 [1.98]	0.25 [1.79]	0.40 [3.09]	0.37 [2.78]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ACI Quintiles					ACI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	363	364	367	367	364	38	42	46	45	40	
	(2)	112	112	113	113	112	76	79	81	82	79	
	(3)	78	79	79	78	78	138	142	142	141	140	
	(4)	68	68	68	68	67	308	314	318	319	316	
(5)	61	62	61	62	62	2151	2252	2585	3043	2425		

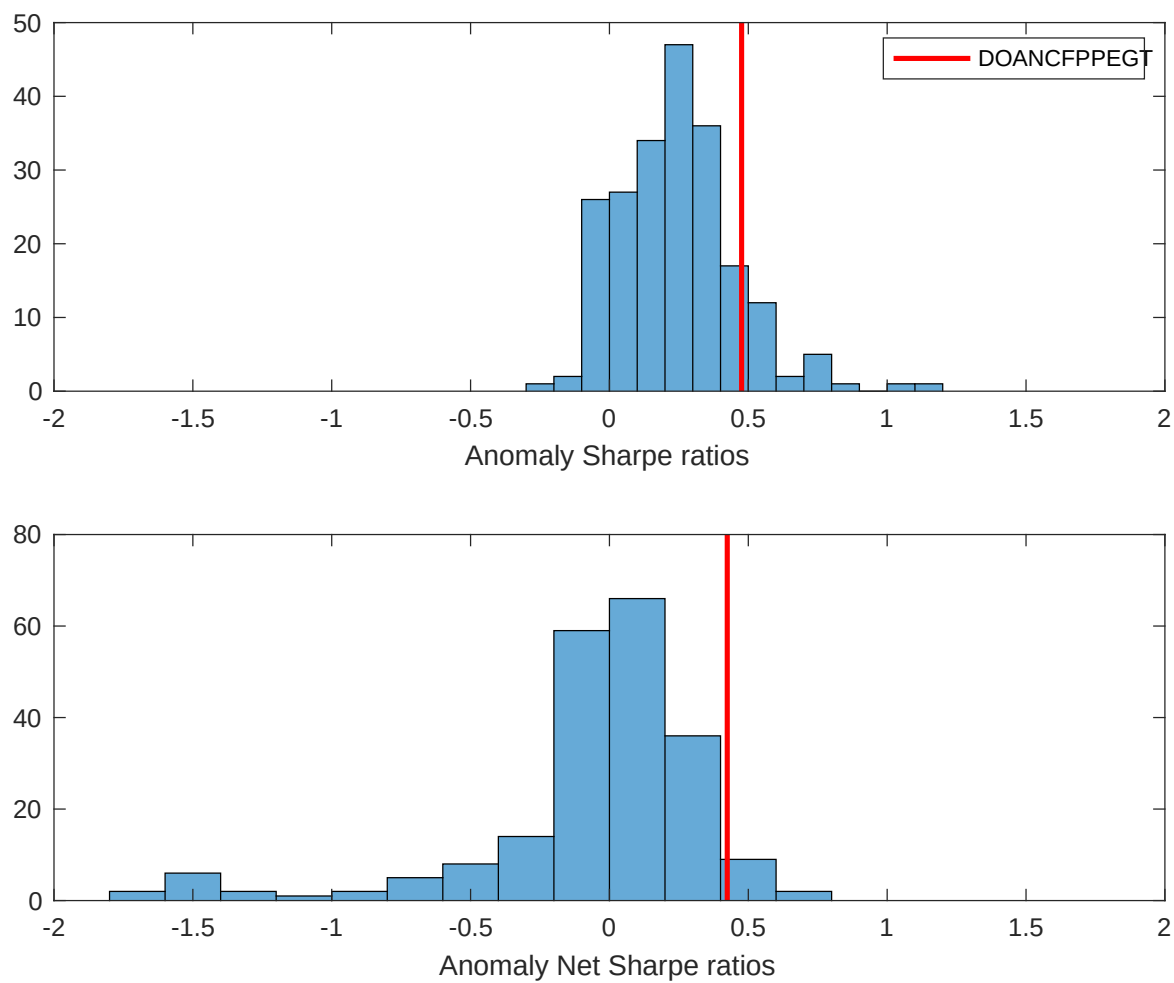


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ACI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

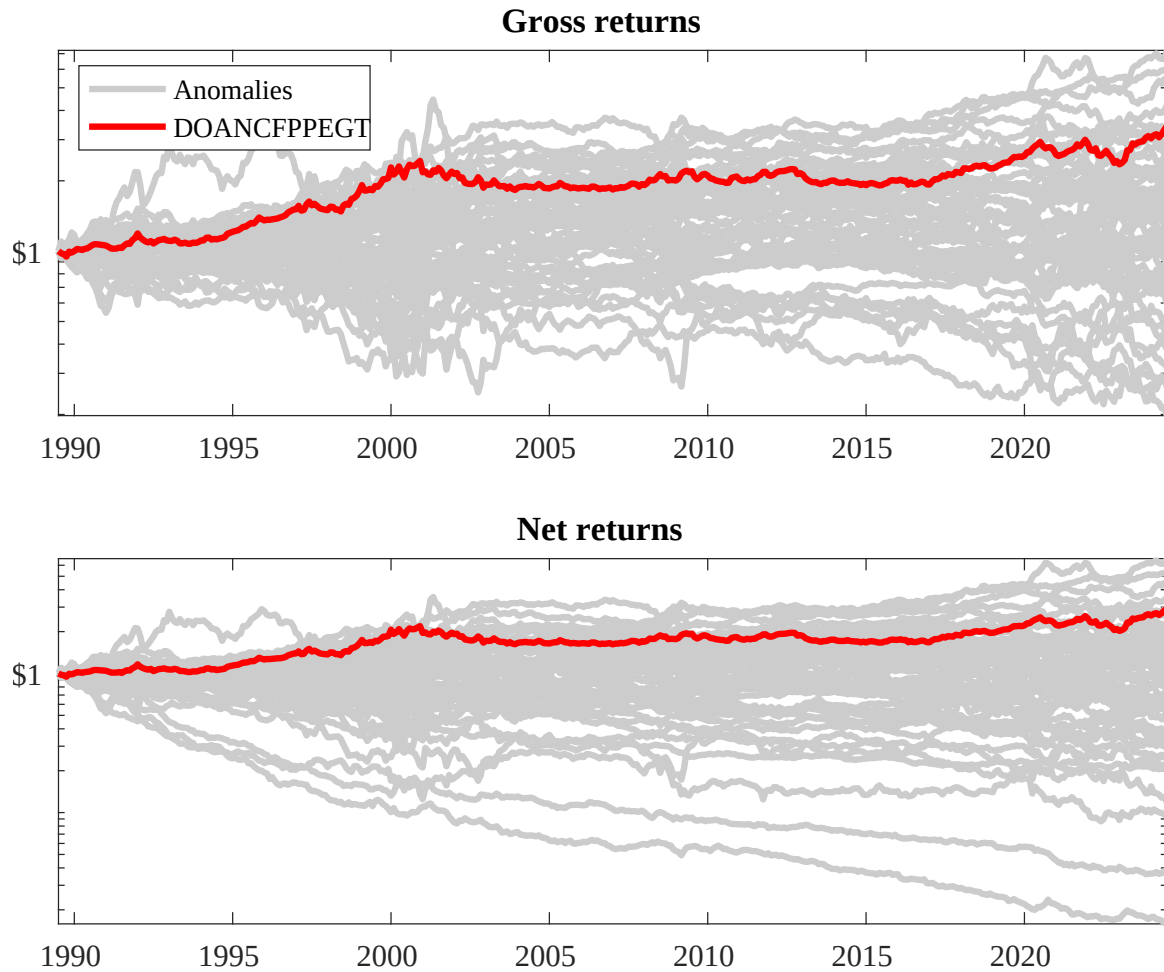
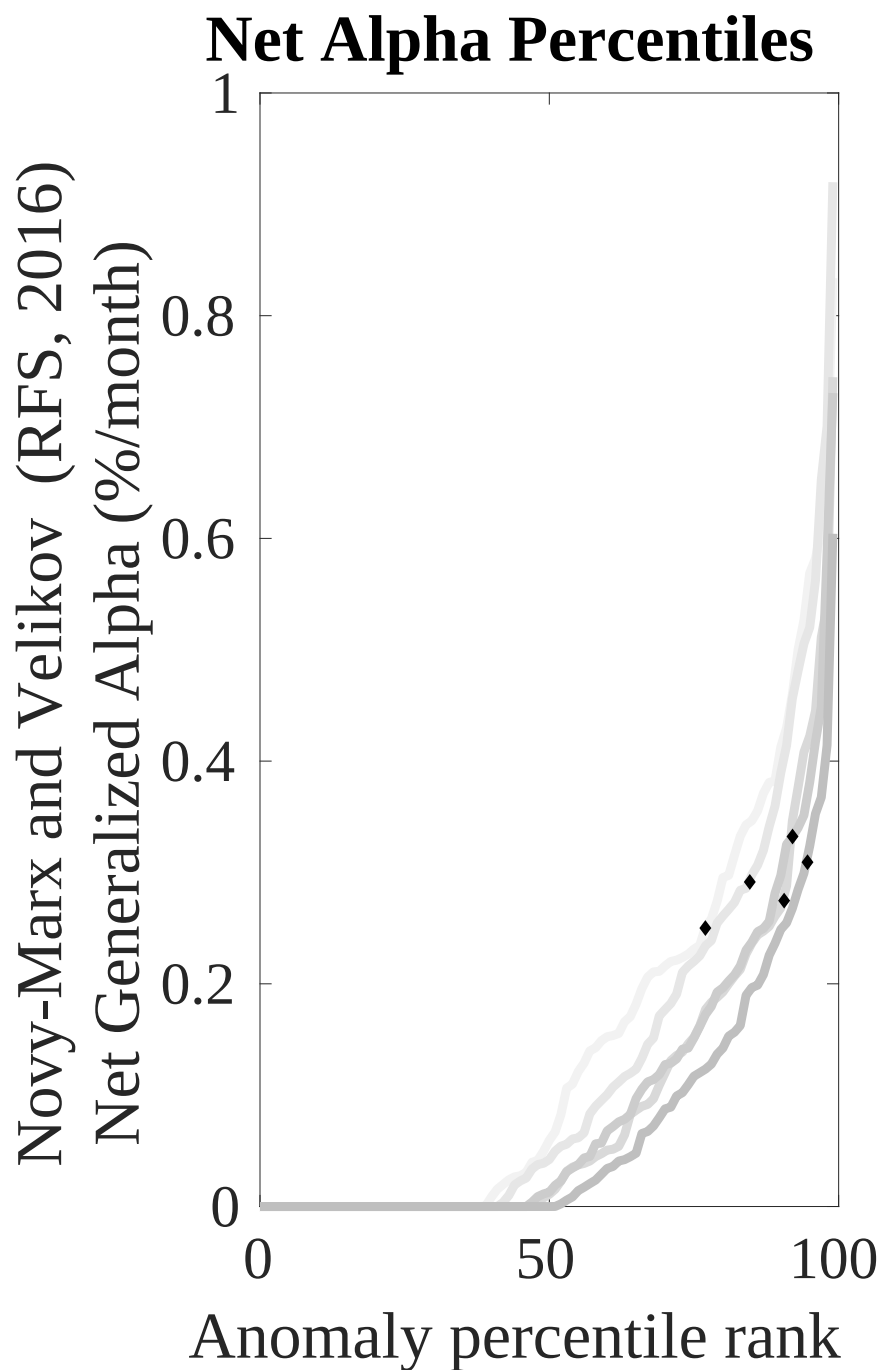
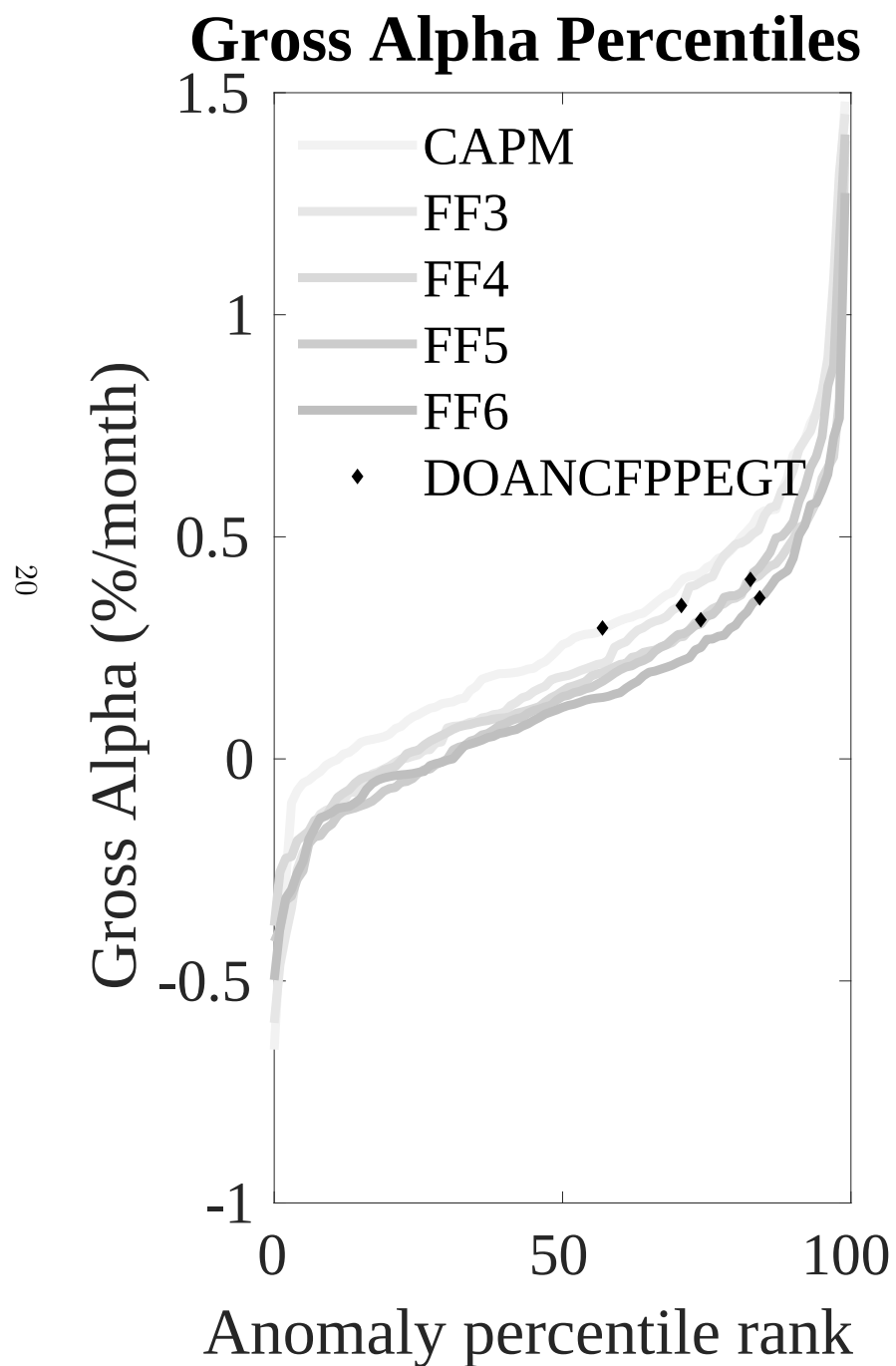


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ACI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



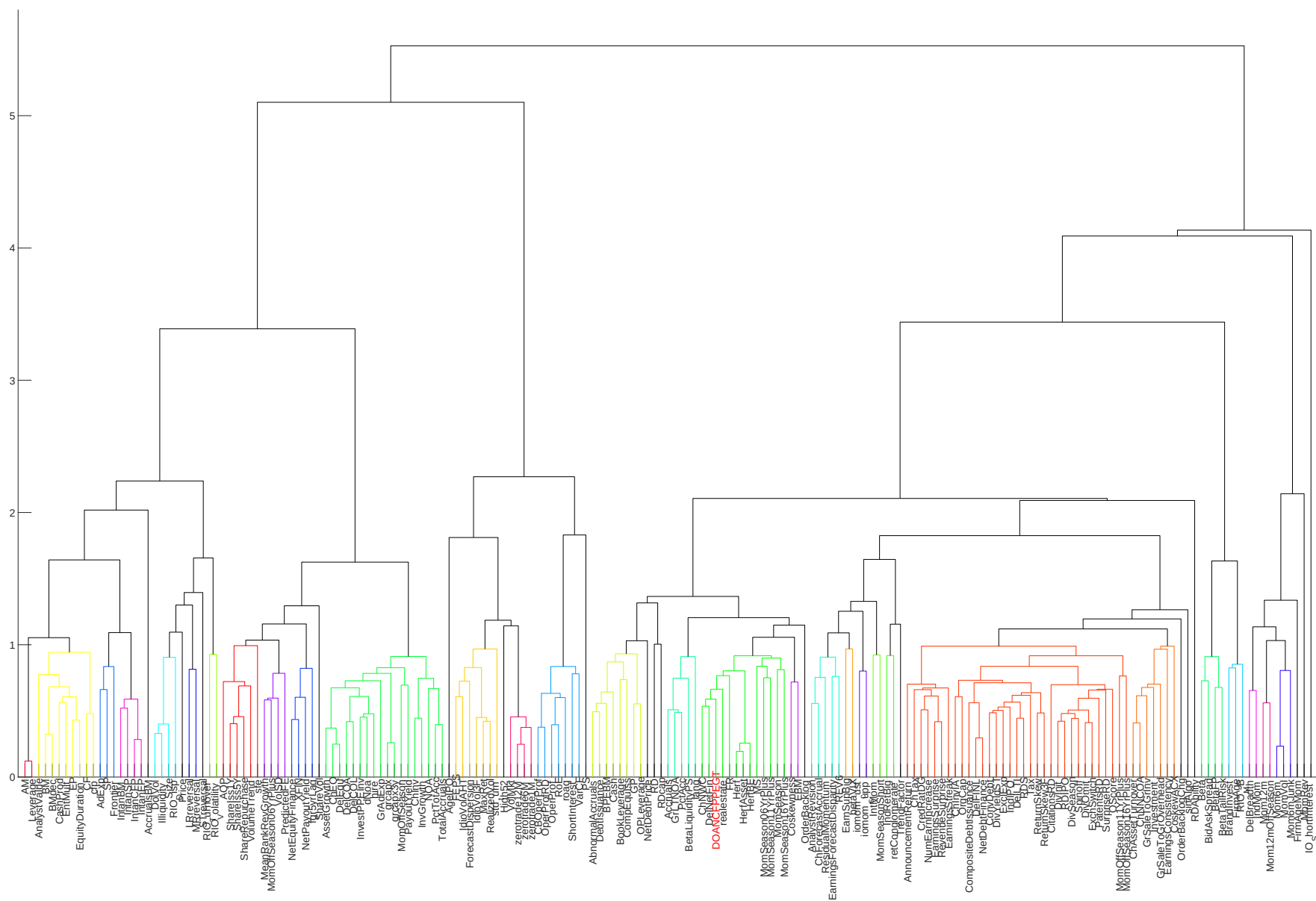


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

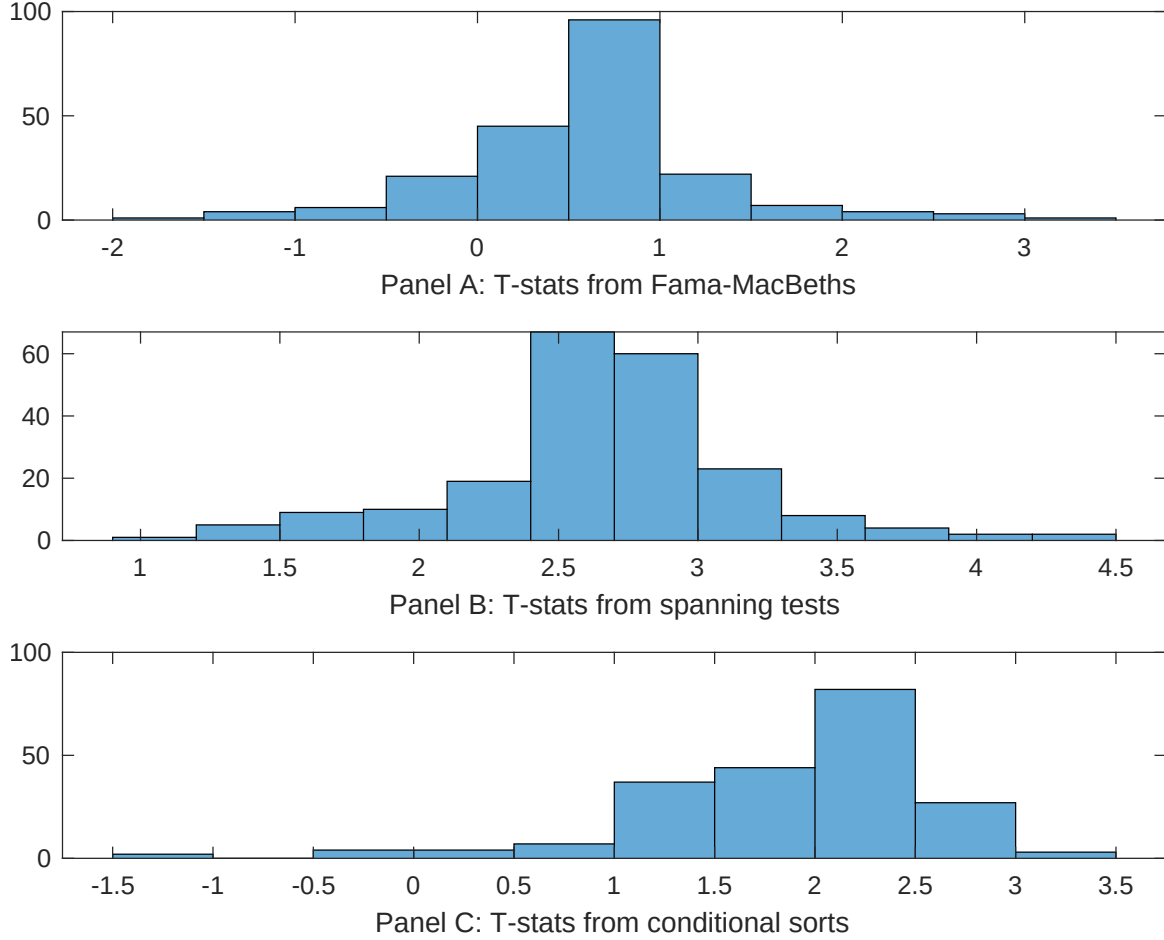


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ACI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ACI. Stocks are finally grouped into five ACI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ACI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.12 [3.97]	0.17 [5.65]	0.12 [3.94]	0.11 [3.84]	0.11 [3.64]	0.13 [4.28]	0.12 [4.53]
ACI	0.26 [1.26]	0.11 [0.51]	0.16 [0.78]	0.41 [0.17]	0.46 [0.00]	0.96 [0.47]	0.35 [0.17]
Anomaly 1	0.13 [4.51]						0.92 [2.13]
Anomaly 2		0.48 [6.27]					-0.25 [-0.30]
Anomaly 3			0.21 [4.41]				0.75 [1.40]
Anomaly 4				0.58 [0.19]			-0.92 [-0.27]
Anomaly 5					0.95 [3.11]		0.78 [2.42]
Anomaly 6						0.87 [7.58]	0.52 [5.03]
# months	420	420	420	420	415	420	415
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ACI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ACI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.37 [3.75]	0.36 [3.64]	0.25 [2.64]	0.27 [2.79]	0.35 [3.43]	0.36 [3.72]	0.19 [2.00]
Anomaly 1	-18.31 [-3.62]						-11.83 [-1.38]
Anomaly 2		-7.96 [-1.48]					14.74 [1.82]
Anomaly 3			-34.74 [-7.99]				-26.85 [-5.60]
Anomaly 4				27.06 [6.08]			19.42 [4.29]
Anomaly 5					0.95 [0.24]		1.84 [0.50]
Anomaly 6						-22.95 [-4.46]	-9.36 [-1.40]
mkt	0.04 [0.02]	0.07 [0.03]	1.09 [0.47]	-2.51 [-1.03]	0.55 [0.22]	0.12 [0.05]	-1.01 [-0.44]
smb	-8.06 [-2.30]	-7.37 [-2.07]	-10.90 [-3.27]	-9.57 [-2.80]	-7.87 [-2.17]	-6.79 [-1.95]	-12.35 [-3.69]
hml	-8.60 [-2.00]	-9.78 [-2.25]	6.56 [1.44]	-6.83 [-1.63]	-10.61 [-2.07]	-8.60 [-2.02]	4.91 [0.95]
rmw	11.93 [2.69]	13.10 [2.92]	13.67 [3.27]	16.35 [3.76]	12.09 [2.59]	12.39 [2.82]	13.20 [3.06]
cma	-36.71 [-4.58]	-47.68 [-5.93]	-36.92 [-5.87]	-49.02 [-8.02]	-55.00 [-8.55]	-27.87 [-3.22]	-26.93 [-3.10]
umd	6.40 [2.96]	6.51 [2.96]	4.55 [2.22]	5.12 [2.42]	6.84 [2.42]	5.74 [2.67]	4.43 [1.69]
# months	420	420	420	420	416	420	416
$\bar{R}^2(\%)$	38	36	45	41	36	39	47

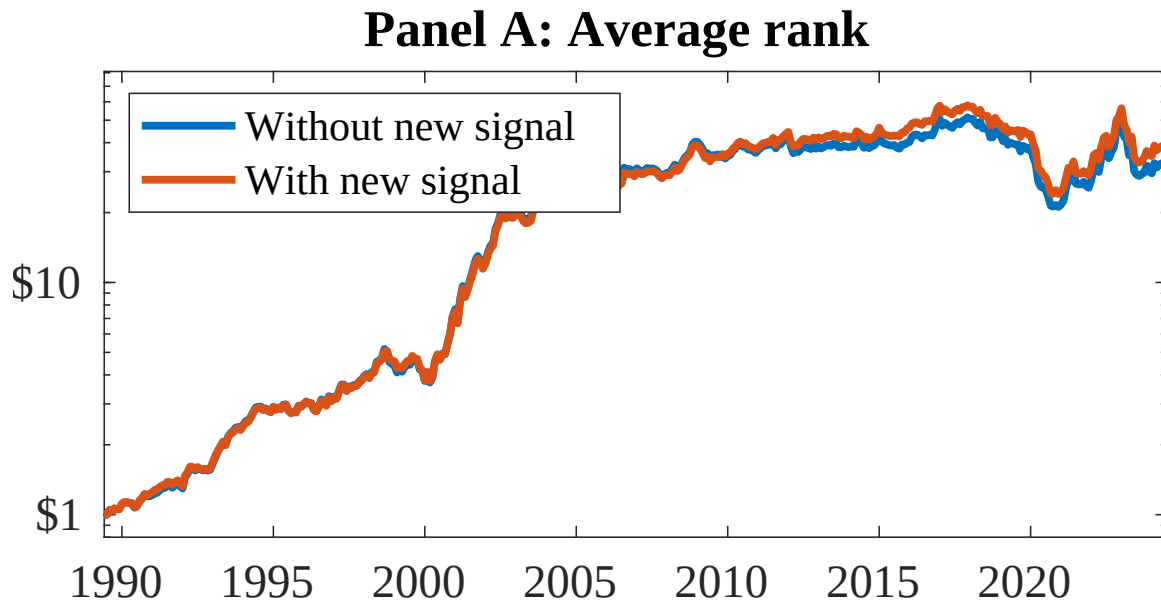


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ACI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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